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Analytical Report 3
Pixelizing Societies from Space
April 24, 2026

Proposed Intervention

Mexico City has experienced rapid urban expansion over the past two decades, with growth increasingly extending into peri-urban areas characterized by mixed vegetation, agricultural land, and low-density settlements. The predictive model developed in Report 2 suggests that future urbanization will continue to concentrate along the urban fringe, particularly in areas with high accessibility and relatively low topographic constraints. These projected patterns raise several environmental concerns, including intensified urban heat island effects, reduced surface permeability leading to flooding, and uneven access to green space across newly developing neighborhoods.

This report proposes a targeted green infrastructure strategy that prioritizes parks and permeable green spaces in areas most likely to urbanize, aiming to guide expansion toward more sustainable outcomes.

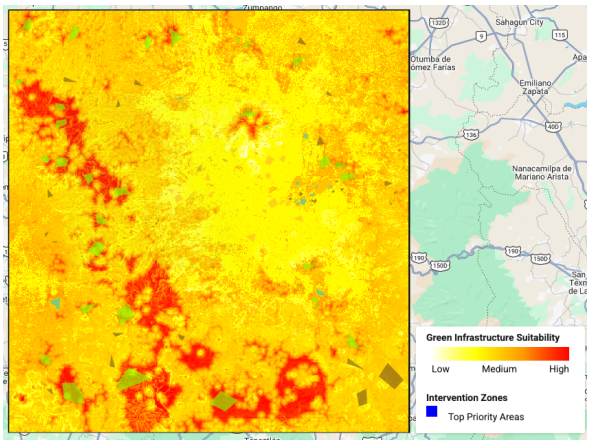
The intervention seeks to reduce environmental risk while improving livability by mitigating heat, enhancing stormwater absorption, and providing accessible public space. By aligning with predicted growth areas, it directs limited planning resources to locations with the greatest long-term impact.

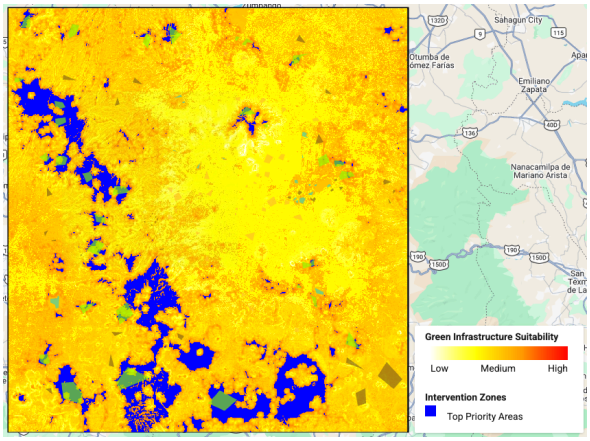
Optimization Results

To identify priority locations, a multi-criteria suitability analysis was conducted using spatial variables from the predictive model and environmental data, combining development likelihood with key physical factors.

Four criteria were used: urban growth probability (40%), distance from existing green space (30%, using vegetation as a proxy), slope (20%), and current land cover (10%, prioritizing non-urban areas). These variables were normalized and combined into a weighted suitability index, producing a continuous surface of relative suitability for green infrastructure investment. Intervention zones were defined as the top 8% of suitability values, corresponding to a threshold of approximately 0.44. This balances selectivity with spatial coverage, ensuring targeted yet meaningful intervention.

The resulting map shows clusters of high-priority zones along the urban periphery, where predicted growth overlaps with limited green space. These areas offer opportunities for early intervention, allowing planners to shape development before it becomes fixed and supporting more targeted, place-based strategies.





Critical Evaluation

The proposed green infrastructure strategy offers clear environmental and social benefits. Increasing permeable surfaces in high-growth areas can reduce flood risk, a major concern in CDMX given its drainage challenges and extreme precipitation. Expanding vegetation can also mitigate urban heat island effects, improving thermal comfort and potentially lowering energy demand. Socially, new parks and green spaces can enhance quality of life and improve access to public amenities. However, the intervention involves important trade-offs. Allocating land for green infrastructure may conflict with housing and commercial development in a rapidly growing city. Land acquisition costs and governance constraints may also limit feasibility. Additionally, new green amenities could increase property values, raising concerns about displacement and uneven distribution of benefits.

Several limitations should be acknowledged. The model relies on physical and spatial variables and does not incorporate detailed socio-economic data, such as population density or income, which may limit its ability to capture inequality or community need. Vegetation was also used as a proxy for green space, which may not reflect access to public parks. Finally, the predictive model is inherently uncertain, and future development may differ from projections. Despite these limitations, the analysis demonstrates how predictive spatial modeling can support more proactive urban planning. By identifying high-impact locations for green infrastructure, CDMX can better manage future growth and move toward more sustainable and resilient development.

Additional

AI tools were used to help interpret data, develop Google Earth Engine code, and assist in structuring the written analysis.
https://code.earthengine.google.com/f0074bde701c88ae8f506393d2856579